



**POLITECNICO**  
MILANO 1863

# High Performance Scientific Computing for Aerospace Project

15th October 2025 update

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- Defined the data structures needed.
- Separation of roles and classes.
- High-level implementation of the code.

- Computation of grid.
- Input values  $\kappa$ ,  $\nu$ , initial conditions handling.
- Establish the order of nodes to maximize performance.
- Potentially excessive memory usage.

- Optimization of the storage in data structures.
- Complete the definition of the data structures for  $\beta$  and  $\gamma$ .
- Decide how to deal with input data  $\kappa$ ,  $\nu$ , initial conditions (if they are a given function or a set of data).
- Discuss about the parallelization strategy for the solution of linear systems.
- Actual implementation of code.